Bilirubin

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Identity Sheet

Property	Value
Name	Bilirubin
Other names	Unconjugated (indirect) bilirubin; Conjugated (direct) bilirubin = bilirubin diglucuronide
Chemical formula	$C_{33}H_{36}N_4O_6$
Molecular weight	584.66 g/mol
CAS number	635-65-4
PubChem CID	5280352
Appearance	Yellow-orange pigment
Solubility	Unconjugated: water-insoluble (bound to albumin); Conjugated: water-soluble

Structural formula:

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Linear tetrapyrrole (open-chain porphyrin derivative):
Pyrrole-CH=Pyrrole-CH2-Pyrrole-CH=Pyrrole
(with vinyl, methyl, propionic acid side chains)
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Bilirubin is the yellow breakdown product of heme degradation. Only **conjugated (direct) bilirubin** appears in urine because unconjugated bilirubin is tightly bound to albumin and is not filtered by the glomerulus. Bilirubinuria is an early sign of hepatobiliary disease.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Urobilinogen	$C_{33}H_{44}N_4O_6$	592.73	Reduced form of bilirubin by gut bacteria; colourless; normally in urine
Biliverdin	$C_{33}H_{34}N_4O_6$	582.65	Green precursor of bilirubin; not found in urine
Stercobilin	$C_{33}H_{46}N_4O_6$	594.74	Final faecal pigment; gives brown colour to stool

Medical Information

Origin

Endogenous: Heme Degradation Bilirubin is produced from heme catabolism (75-85% from senescent RBC haemoglobin, 15-25% from other haemoproteins). Pathway: Heme -> biliverdin (heme oxygenase) -> bilirubin (biliverdin reductase). Unconjugated bilirubin is transported bound to albumin to the liver, where UDP-glucuronosyltransferase (UGT1A1) conjugates it with glucuronic acid -> conjugated bilirubin (water-soluble), excreted in bile.

Daily production: ~250-350 mg bilirubin.

Exogenous No significant dietary source. Bilirubin is exclusively endogenous.

Primary & Secondary Biological Roles

Primary role: - **Waste product:** Bilirubin is an excretory end-product of heme metabolism.

Secondary roles: - **Antioxidant:** Bilirubin is a potent scavenger of peroxyl radicals; mildly elevated levels may be cardioprotective (Gilbert syndrome). - **Clinical marker:** Serum bilirubin levels indicate liver function, haemolysis, and biliary obstruction.

Catabolism and Elimination Pathway

- Conjugated bilirubin is excreted in bile into the intestine.
- Gut bacteria reduce conjugated bilirubin -> urobilinogen -> stercobilin (faecal colour).
- ~10-15% of urobilinogen is reabsorbed (enterohepatic circulation), re-excreted in bile or filtered by kidneys into urine.
- Conjugated bilirubin can be filtered by the glomerulus if serum levels are elevated -> bilirubinuria.
- Unconjugated bilirubin is not excreted in urine (albumin-bound).

Expression in Humans

Normal Levels

Compartment	Reference Range
Total serum bilirubin	0.1-1.2 mg/dL (2-21 umol/L)
Direct (conjugated)	0-0.3 mg/dL
Indirect (unconjugated)	0.1-0.8 mg/dL
Urine bilirubin	Negative (absent or <0.02 mg/dL)
Urine urobilinogen	0.1-1.0 Ehrlich units (trace amounts normal)

Factors Influencing Levels **Bilirubinuria (positive urine bilirubin):** - Obstructive jaundice (gallstones, pancreatic head tumour) - Hepatocellular disease (hepatitis, cirrhosis) - Drug-induced cholestasis - Dubin-Johnson syndrome (rare conjugated hyperbilirubinaemia)

Absent urine bilirubin (normal or haemolytic): - Normal state - Haemolytic jaundice (unconjugated bilirubin elevated but not filtered) - Gilbert syndrome (unconjugated; no bilirubinuria)

Associated Pathologies

Condition	Bilirubin Pattern	Key Symptoms
Obstructive jaundice	Positive urine bilirubin + absent urobilinogen	Jaundice, dark urine, pale stools, pruritus

Condition	Bilirubin Pattern	Key Symptoms
Hepatitis	Positive urine bilirubin + variable urobilinogen	Jaundice, RUQ pain, fatigue
Haemolytic jaundice	Negative urine bilirubin + increased urobilinogen	Anaemia, jaundice, splenomegaly
Gilbert syndrome	Negative urine bilirubin	Mild intermittent jaundice; benign; ~5% population

Presence in Urine

Should it be normally present? No – urine bilirubin should be absent. Its presence (bilirubinuria) always indicates pathology (conjugated hyperbilirubinaemia).

Normal urinary levels: Negative (<0.02 mg/dL).

Form in urine: Conjugated bilirubin (bilirubin diglucuronide), water-soluble form.

Pathological significance:

Urinary Bilirubin	Possible Causes	Prevalence
Positive	Obstructive or hepatocellular jaundice	Varies; gallstones ~10-15% adults
Negative	Normal; or haemolytic jaundice (unconjugated)	–

Solubility: Conjugated bilirubin is water-soluble. Unconjugated is insoluble (albumin-bound).

Detection in Urine

Available Clinical Assays

1. **Dipstick diazo reaction:**
   - **Principle:** Conjugated bilirubin reacts with diazonium salt (2,4-dichloroaniline or similar) -> azobilirubin (pink-tan colour).
   - **Detection:** Colour change on pad.
   - **LOD:** ~0.5 mg/dL (~8.5 umol/L).
   - **Advantages:** Rapid, included on standard dipstick.
   - **Disadvantages:** Light degrades bilirubin -> false negatives; insensitive.
2. **Ictotest (confirmatory tablet test):**
   - **Principle:** Bilirubin adsorbs onto special mat; diazo tablet reagent -> blue-purple colour.
   - **Detection:** Visual colour on mat.
   - **LOD:** ~0.05-0.1 mg/dL (more sensitive than dipstick).
   - **Advantages:** Confirmatory; Siemens Ictotest tablets.
3. **Total bilirubin by diazo (Jendrassik-Grof, lab):**
   - **Principle:** Bilirubin + diazotised sulfanilic acid -> azobilirubin.
   - **Detection:** Absorbance at **540-560 nm**.
   - **LOD:** ~0.1 mg/dL.

Optimal Urine Type for Measurement

- **Fresh specimen** essential – bilirubin is photolabile (decomposes rapidly in light).
- Protect from light (foil-wrapped container).
- Analyse within **1 hour** of collection.
- Random specimen adequate.

Actual Gold Standard

Ictotest tablet is the confirmatory urine bilirubin test. For serum, the **Jendrassik-Grof diazo method** at 540 nm is the reference.

Interferences in Measurement

Interference	Effect	Mechanism
Light exposure	False negative	Photodecomposition of bilirubin
Ascorbic acid	False negative	Reduces diazo reaction
Nitrite (high)	False negative	Oxidises bilirubin
Chlorpromazine	False positive	Produces coloured metabolites
Phenazopyridine	Colour masking	Orange dye
Elevated urobilinogen	Minor interference	Slight colour contribution

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis:** Bilirubin absorbs strongly at **453 nm** (epsilon ~55,000 M-1cm-1 in chloroform; shifts in aqueous solution). Direct spectrophotometry at 453 nm possible if concentrated enough.
  - LOD: ~0.05 mg/dL in clean solutions.
  - In urine matrix: interfered by other chromogens.
- **NIR:** No useful bilirubin absorption.

Fluorescence Detection

- Bilirubin has very weak native fluorescence (quantum yield <0.01%) due to rapid internal conversion.
- **Zinc bilirubin complex:** Zn-bilirubin fluoresces (Ex 430 / Em 530 nm). LOD: ~0.01 mg/dL.
- **Bilirubin oxidase + fluorescent product:** Enzymatic oxidation produces biliverdin (weakly fluorescent).
- Not commonly used for urine bilirubin.

Raman Detection

Peak (cm-1)	Assignment
~1615	C=C stretching (vinyl groups)
~1575	Pyrrole ring stretching
~1340	C-N stretching
~1270	C-H bending

- Bilirubin has characteristic Raman spectrum but concentrations in urine are too low for conventional Raman.
- SERS: Potential LOD ~0.01 mg/dL with Au nanoparticles. Limited studies.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1690	C=O stretch (lactam)	Strong
~1620	C=C stretch	–
~1570	Pyrrole ring	–
~3300	N-H stretch	Broad

- ATR-FTIR: Not sensitive enough for normal or mildly elevated urinary bilirubin.

Voltammetry Detection

- Bilirubin is electrochemically oxidised at **+0.1-0.4 V vs Ag/AgCl** on carbon electrodes.
- LOD: ~0.01-0.1 mg/dL on modified electrodes (graphene, MWCNTs).
- Interference from ascorbic acid and uric acid (similar oxidation potentials).
- **MIP sensors:** Molecularly imprinted polymers for bilirubin. LOD: ~0.005 mg/dL.

Other Detection Technologies

1. **Transcutaneous bilirubinometry:** Non-invasive skin reflectance at 460/550 nm for neonatal jaundice screening. Not for urine.
2. **Smartphone-based dipstick reading:** Camera analysis of diazo pad colour. LOD: similar to visual (~0.5 mg/dL).
3. **Paper-based diazo assay:** Low-cost diazo-impregnated paper with smartphone readout.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Dipstick (diazo)	~0.5 mg/dL	Pink-tan colour	None	Standard, rapid
Ictotest	~0.05 mg/dL	Blue-purple	Mat adsorption	More sensitive
Jendrassik-Grof	~0.1 mg/dL	540-560 nm	Reagent	Quantitative
UV-Vis (453 nm)	~0.05 mg/dL	453 nm	None	Direct
Zn-bilirubin fluorescence	~0.01 mg/dL	Ex 430/Em 530 nm	Zn addition	Sensitive
Voltammetry	~0.01 mg/dL	+0.1-0.4 V	Buffer	Sensitive
MIP electrochemical	~0.005 mg/dL	DPV	Buffer	Selective
Raman/FTIR	Not practical	–	–	Concentrations too low

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